PhD Math Camp

Calculus I: Derivatives

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Outline

Derivative

Introduction

- In social sciences, understanding the rate of change in a function $f(x) = y$, is essential when studying how an independent or explanatory variable x influences a dependent or outcome variable y.
- We generally consider two types of rates of change:
    1. **Average Rate of Change**: Measures how much a function changes between two points.
    2. **Instantaneous Rate of Change**: Measures the rate of change at a specific point.

Average Rate of Change

- The average rate of change of a function between two points is the change in the function's output divided by the change in the input.
- Visually, this is represented as the slope of the secant line connecting two points on the function's graph.

Formula for Average Rate of Change

- Consider a function $f(x)$ with two points $(a, f(a))$ and $(b, f(b))$.
- The average rate of change is given by:

$$\text{Average Rate of Change} = \frac{f(b) - f(a)}{b - a}$$

- This formula computes the slope of the secant line, quantifying the average change over the interval.

Example of Average Rate of Change

- For the function $f(x) = x^2$, calculate the average rate of change from $x = 2$ to $x = 5$:

$$f(2) = 4, \quad f(5) = 25$$

$$\text{Average Rate of Change} = \frac{25 - 4}{5 - 2} = 7$$

- The function increases by 7 units on average for each unit increase in x over this interval.

Transition to Instantaneous Rate of Change

- The instantaneous rate of change examines the behavior of a function at a specific point.
- This is equivalent to finding the slope of the tangent line to the curve at that point.
- As the interval shrinks, the secant line approaches the tangent line.

Secant Line and Slope

- The slope of the secant line between $(a, f(a))$ and $(b, f(b))$ is:

$$\text{Slope of Secant Line} = m = \frac{f(b) - f(a)}{b - a}$$

- The equation of the secant line is:

$$y - f(a) = m \cdot (x - a)$$

$$y - f(a) = \left(\frac{f(b) - f(a)}{b - a}\right)(x - a)$$

Deriving the Tangent Line Equation

- The slope of the tangent line at $x = a$ is obtained by taking the limit as b approaches a:

$$m = \lim_{b \to a} \frac{f(b) - f(a)}{b - a} = f'(a)$$

- The equation of the tangent line at $(a, f(a))$ is:

$$y - f(a) = f'(a)(x - a)$$

- Simplifying:

$$y = f'(a)(x - a) + f(a)$$

Explore More: Interactive Tools and Resources

- **GeoGebra - Secant and Tangent Lines**: Explore the concepts of secant and tangent lines with this interactive tool.
- **Animated Mathematics - Differentiation from First Principles**: Visualize differentiation using first principles.
- **Zooming in on a Tangent Line**: Understand how zooming in on a curve shows the tangent line using this interactive tool.
- **Desmos - Interactive Derivative Explorer**: Experiment with derivatives using the Desmos calculator.
- **Desmos - Secant and Tangent Line Explorer**: Further explore the relationship between secant and tangent lines with this Desmos calculator.

Understanding Derivatives

- The derivative $f'(a)$ represents the instantaneous rate of change of the function at $x = a$.
- It provides insight into how the function behaves at that specific point.
- In social sciences, we use functions to represent relations between variables.
- Thus, derivatives are useful because they tell two things:
  - The sign or direction of association between two variables.
  - The strength of the association between two variables.
  - We can assess these two aspects at different points of the function $f(x) = y$.

Derivatives

- The derivative is the instantaneous rate of change of a function.
- The instantaneous rate: the slope of the tangent line.

Definition

Def. Let f be a function whose domain includes an open interval containing the point x. The derivative of f at x is given by:

$$\frac{d}{dx}f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Proof of the Derivative of x^2 Using Limits

Thm. The derivative of $f(x) = x^2$ is $\frac{d}{dx}[x^2] = 2x$.

Proof.

$$\begin{aligned}
\frac{d}{dx}[x^2] &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\
&= \lim_{h \to 0} \frac{2xh + h^2}{h} \\
&= \lim_{h \to 0} (2x + h) \\
&= 2x
\end{aligned}$$

∎

Conclusion: The derivative of x^2 is $2x$.

Proof of the Power Rule for Derivatives

Thm. If $f(x) = x^n$ where n is a positive integer, then $\frac{d}{dx}[x^n] = nx^{n-1}$.

Proof.

$$
\begin{aligned}
\frac{d}{dx}[x^n] &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{(x+h)^n - x^n}{h} \\
&= \lim_{h\to 0} \frac{\sum_{k=0}^{n} \binom{n}{k} x^{n-k} h^k - x^n}{h} \\
&= \lim_{h\to 0} \frac{x^n + nx^{n-1}h + \frac{n(n-1)}{2} x^{n-2} h^2 + \cdots + h^n - x^n}{h} \\
&= \lim_{h\to 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2} x^{n-2} h^2 + \cdots + h^n}{h} \\
&= \lim_{h\to 0} \left(nx^{n-1} + \frac{n(n-1)}{2} x^{n-2} h + \cdots + h^{n-1} \right) = nx^{n-1}
\end{aligned}
$$

∎

Notation

- **Leibniz notation:** $\frac{d}{dx} f(x)$
- **Lagrange's prime notation:** $f'(x)$
- **Euler's notation:** $D_x f(x)$
- Read as "the derivative of f of x with respect to x."

Existence Condition

- If f is differentiable at $x = c$, then f is continuous at $x = c$.
- **Note:** The converse does not hold.

Some Rules

- **Power rule:**
$$\frac{d}{dx}x^n = nx^{n-1}$$
- The derivative is a linear operator.
- **Sum rule:** $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$
- **Multiply by constant rule:** $[kf(x)]' = kf'(x)$
- **Product rule:**
$$[f(x) \cdot g(x)]' = f'(x)g(x) + f(x)g'(x)$$
- **Quotient rule:**
$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Exercise

1. $\frac{d}{dx}x^2$
2. $\frac{d}{dx}3x^3$
3. $\frac{d}{dx}x$
4. $\frac{d}{dx}5$
5. $\frac{d}{dx}\sqrt{x}$
6. $\frac{d}{dx}(3x^4 + 3x)$
7. $\frac{d}{dx}(3x^2 - 3)(4x^3 - 2x)$

Chain Rule

- Differentiate nested functions:

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

- **Examples:**
  1. $\frac{d}{dx}(3x^2 - 3)^3$
  2. Prove the quotient rule by the product rule and the power rule.
  3. Prove $\frac{d}{dx}\left[f(g(x)) \cdot h(x)\right] = f'(g(x))g'(x)h(x) + f(g(x))h'(x)$
  4. $\frac{d}{dx}(3x^2 - 3)^2(4x^3 - 2x)$
- **Exercise:**

$$\frac{d}{dx}(x^2 + x + 2)^2$$

Derivatives of Logarithms and Exponents

- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \log(x) = \frac{1}{x}$
- $\frac{d}{dx} n^x = \log(n) n^x$
- $\frac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$ *(Prove by the chain rule)*
- $\frac{d}{dx} \log(f(x)) = \frac{f'(x)}{f(x)}$ *(Prove by the chain rule)*

Exercise:

$$\frac{d}{dx} e^{2x^2}$$

Implicit Differentiation Technique

What is Implicit Differentiation?

- **When to Use It:** Implicit differentiation is used when a function is not explicitly solved for one variable in terms of another (e.g., y is not isolated).
- **How It Works:** Differentiate both sides of an equation with respect to x, treating y as a function of x (i.e., apply the chain rule when differentiating terms involving y).

Ex.

$$\text{Given } 3x^2 + \log(3 + 4y) = 25, \text{ find } \frac{dy}{dx}.$$

Logarithmic Differentiation Technique

Why Use Logarithmic Differentiation?

- **Simplifies Finding Extrema:** For complex functions, direct differentiation can be challenging. Logarithms transform products and powers into sums and simpler forms, making derivatives easier to calculate for identifying maxima or minima.
- **Eases the Differentiation Process:** By applying logarithms and differentiating implicitly, we can simplify the derivative process, especially for functions involving products or powers.

Ex.

$$y = \left(3x^2 - 3\right)^{\frac{1}{3}} (4x - 2)^{\frac{1}{4}} \left(x^2 + 1\right)^{\frac{1}{2}}$$

Mean Value Theorem

Informal: The theorem states that the derivative of a continuous and differentiable function must attain the function's average rate of change (in a given interval). For instance, if a car travels 100 miles in 2 hours, then it must have had the exact speed of 50 mph at some point in time.

Thm. Suppose that a function f is continuous on the closed interval $[a, b]$, and differentiable on the open interval (a, b). Then, $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Second (and Higher Order) Derivatives

Second derivative: Rate of change of the rate of change, i.e., the rate of change of the slope of a tangent line.

$$f''(x), \quad \frac{d^2}{dx^2} f(x)$$

We can also do nth derivative:

$$\frac{d^n}{dx^n} f(x)$$

Example:

$$\frac{d^2}{dx^2} 2x^3$$

Exercise:

$$\frac{d^2}{dx^2} e^{2x^2}$$

Taylor Series Expansion

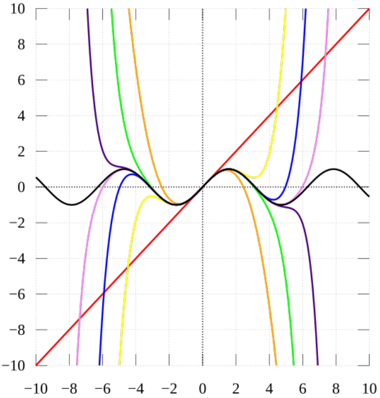
A Taylor series is a way to represent common functions as infinite series (a sum of infinite elements) of the function's derivatives at some point a.

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Taylor Expansion: Interactive Demo

Taylor Series Expansion



As the degree of the Taylor polynomial rises, it approaches the correct function. This image shows $\sin x$ *and its Taylor approximations by polynomials of degree 1, 3, 5, 7, 9, 11, and 13 at* $x = 0$*. (from Wikipedia)*

Partial Derivatives

Example: $f(x, y) = x^3y^4 + e^x - \log y$ (from Harvard Math Prefresher for Political Scientists)

$$\frac{\partial}{\partial x} f(x, y)$$

$$\frac{\partial}{\partial y} f(x, y)$$

$$\frac{\partial^2}{\partial x^2} f(x, y)$$

$$\frac{\partial^2}{\partial x \partial y} f(x, y)$$

$$\frac{\partial^2}{\partial y \partial x} f(x, y)$$

More Exercises

Find the partial derivative with respect to z:

1. $f(x, y, z) = 3x + 4y + 5z + 6$
2. $f(x, y, z) = 11z + 3x^2y + 5x^2z + 7z^2y$

Differentiate the following:

1. $f(x) = e^{x - \log(x) + 5}$
2. $f(x) = \left(x^3 + 2\right) \log\left(x^4 - 5x + 3\right)$
3. $f(x) = \left(\frac{x^2+1}{x+1}\right)^2$

Optimization

Concavity of a Function

- **Strict Concave:** A function f is strictly concave if $\forall x_1, x_2$ in its domain and any weight $\lambda \in (0, 1)$

$$f(\lambda x_1 + (1-\lambda)x_2) > \lambda f(x_1) + (1-\lambda)f(x_2)$$

- **Strict Convex:** A function f is strictly convex if $\forall x_1, x_2$ in its domain and any weight $\lambda \in (0, 1)$

$$f(\lambda x_1 + (1-\lambda)x_2) < \lambda f(x_1) + (1-\lambda)f(x_2)$$

- **Second Derivative Test of Concavity:**

$$f''(x) < 0 \Rightarrow \text{Concave}$$

$$f''(x) > 0 \Rightarrow \text{Convex}$$

Why Concavity Matters in Statistical Modeling

The Core Idea: Statistics as Optimization

Many common statistical methods, like regression or maximum likelihood estimation, are fundamentally *optimization problems*. We are trying to find the best parameters for our model by either:

- **Maximizing** a "goodness-of-fit" function (like a Likelihood Function).
- **Minimizing** an "error" function (like a Loss Function).

What's the Role of Concavity and Convexity? If the function we optimize is *concave*, any local maximum is the global maximum. If it is *convex*, any local minimum is the global minimum. This guarantees uniqueness and stability of solutions, making optimization tractable.

Application: Risk Aversion Utility

The standard assumption of a utility function (e.g., Alt and Iverson (2017)):

$$u'(x) > 0, \quad \text{and} \quad u''(x) < 0$$

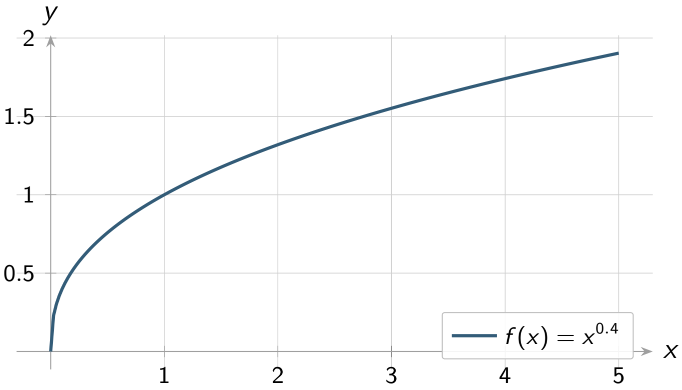
E.g.,

$$u(x) = x^{\alpha}, \quad \text{where} \quad \alpha < 0$$

Takeaway

In game theory and formal modeling, the function that represents the level of satisfaction of an agent, the utility function $u(x)$, given a decision or policy x is often modeled as a globally concave function.

Graph of $f(x) = x^{0.4}$ with $f'(x)$ and $f''(x)$



Example of a risk aversion utility function.

Extrema in One Dimension

First-Order Condition: Formal Definition

First-Order Necessary Condition

Let a function f be defined on an open interval containing a point c. If f has a local extremum (either a local maximum or a local minimum) at $x = c$, and if the derivative $f'(c)$ exists, then it must be that:

$$f'(c) = 0$$

- **Interpretation:** This is a *necessary condition* for an extremum to exist at a point where the function is differentiable. It states that if a point is a peak or a valley, its slope must be zero.
- **Critical Points:** The values of x for which $f'(x) = 0$ or $f'(x)$ is undefined are known as the **critical points** of the function.
- **Important Caveat:** This condition is not sufficient. A point where $f'(x) = 0$ is a *candidate* for an extremum, but it could also be an inflection point (where the curve changes concavity).

Maxima and Minima

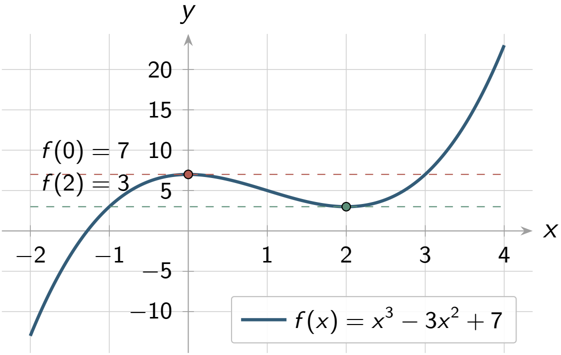
Maxima and minima: They are the high and low points of a function, or the “peaks” and “valleys” in the graph of a function. The value of x that maximizes (minimizes) the function value:

$$x^* = \arg\max_x f(x) \quad (\arg\min_x f(x))$$

Local Maximum (Minimum): x^* is a local maximum (minimum) if $f(x^*) > f(x)$ $(f(x^*) < f(x))$ for all x within some open interval containing x^*.
Global Maximum (Minimum): x^* is a global maximum (minimum) if $f(x^*) > f(x)$ $(f(x^*) < f(x))$ for all x in the domain of f.

First-Order Condition (FOC)

Consider the function: $f(x) = x^3 - 3x^2 + 7$



From the graph, you can see the slopes of all the tangent lines at extrema are 0, i.e., $f'(x) = 0$.

First-Order Condition (FOC)

Ex. Using the FOC find the extrema for this function:

$$f(x) = (3x - 2)^2$$

Second-Order Condition

What is the Second-Order Condition?

- The **second-order condition** refers to the use of the second derivative $f''(x)$ to determine whether a critical point (where $f'(x) = 0$) is a local maximum, minimum, or an inflection point.
- Specifically:
  - If $f''(x) > 0$ at a critical point, the function is **concave up**, indicating a **local minimum**.
  - If $f''(x) < 0$ at a critical point, the function is **concave down**, indicating a **local maximum**.
  - If $f''(x) = 0$, the test is inconclusive, and the point may be an inflection point.

Second-Order Condition

Intuitive Explanation:

- The **second derivative** $f''(x)$ can be understood as the *rate of change of the first derivative* $f'(x)$.
- At an **extrema** (maximum or minimum), the first derivative $f'(x)$ is zero, meaning the slope of the tangent is flat.
- The second derivative tells us how this slope changes around the extrema:
  - A positive $f''(x)$ means the slope $f'(x)$ is increasing, leading to a minimum.
  - A negative $f''(x)$ means the slope $f'(x)$ is decreasing, leading to a maximum.

Inflection Points

Inflection Point

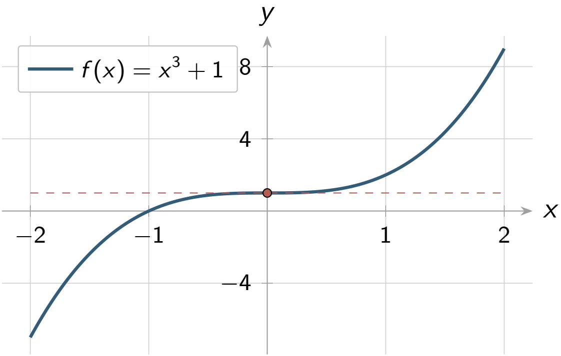
An **inflection point** of a function $f(x)$ is a point on the curve where the concavity of the function changes. This means the function changes from being concave up (convex) to concave down (concave) or vice versa.

Conditions:

- $f''(x) = 0$ or $f''(x)$ does not exist.
- The sign of $f''(x)$ must change around the point.

Inflection Points

Consider the function: $f(x) = x^3 + 1$



Plot of $f(x) = x^3 + 1$

Global Maxima and Minima

1. Globally concave up or concave down functions. If f'' is never zero, then there is at most one critical point. That critical point is a global maximum if $f'' < 0$ and a global minimum if $f'' > 0$.
2. Functions over closed and bounded intervals must have both a global maximum and a global minimum.

Exercise

Find all the extrema for $f(x)$, then use the second order condition to determine if the extrema are a minimum, maximum, or inflection point. Further, determine if the extrema are local or global.

$$f(x) = x^2 - 2x + 17$$

Application 1

The utility that a legislator extracts from a policy in a one-dimensional policy space varies as a quadratic loss function, $U(x)$, of the form

$$U(x) = -a(x - x_0)^2$$

where x_0 represents the ideal location of the legislator and x represents the location of the current policy in the one-dimensional policy space.

1. Prove that the legislator's utility is maximized when the policy is located at $x = x_0$.
2. Is the utility function convex or concave?

Application 2

Given:

- Two independent estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ for a parameter θ.
- A weighted average estimator $\hat{\theta}_w = w\hat{\theta}_1 + (1 - w)\hat{\theta}_2$ where $0 \leq w \leq 1$.

Goal:

- Find the value of w that minimizes the variance of $\hat{\theta}_w$.

Variance of the weighted estimator:

$$\mathrm{Var}(\hat{\theta}_w) = w^2\mathrm{Var}(\hat{\theta}_1) + (1 - w)^2\mathrm{Var}(\hat{\theta}_2)$$

Application 2: Solution

Minimizing the variance:

- Take the derivative of $\mathrm{Var}(\hat{\theta}_w)$ with respect to w:

$$\frac{d}{dw}\left(w^2\mathrm{Var}(\hat{\theta}_1) + (1-w)^2\mathrm{Var}(\hat{\theta}_2)\right) = 0$$

- Simplify and solve for w:

$$2w\mathrm{Var}(\hat{\theta}_1) - 2(1-w)\mathrm{Var}(\hat{\theta}_2) = 0$$

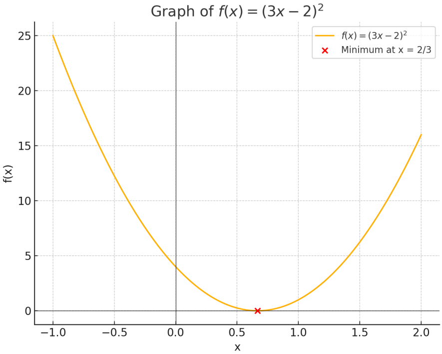
$$w = \frac{\mathrm{Var}(\hat{\theta}_2)}{\mathrm{Var}(\hat{\theta}_1) + \mathrm{Var}(\hat{\theta}_2)}$$

Closed-Form Solution

Consider the minimization problem:

$$\arg\min_x f(x) = (3x-2)^2$$

Closed-Form Minimization: Interactive Demo



Taylor Series Approximation of $\sqrt{2}$

Goal: Approximate the value of $\sqrt{2}$ using a third-order Taylor series expansion around $x = 1$.
Taylor Series Expansion:

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \frac{f'''(a)}{6}(x-a)^3$$

where $f(x) = \sqrt{x}$, $a = 1$, and $x = 2$.
Calculations:

- $f(x) = \sqrt{x}$
- $f'(x) = \frac{1}{2\sqrt{x}}$
- $f''(x) = -\frac{1}{4x^{3/2}}$
- $f'''(x) = \frac{3}{8x^{5/2}}$

Applying the Third-Order Taylor Series and Checking the Approximation

Using the Taylor Series:

$$\sqrt{2} \approx f(1) + f'(1)(2-1) + \frac{f''(1)}{2}(2-1)^2 + \frac{f'''(1)}{6}(2-1)^3$$

$$f(1) = 1 \qquad f'(1) = \frac{1}{2} \qquad f''(1) = -\frac{1}{4} \qquad f'''(1) = \frac{3}{8}$$

$$\sqrt{2} \approx 1 + 0.5 - 0.125 + \frac{3/8}{6} = \frac{23}{16} \approx 1.4375$$

Checking the Approximation:

$$\left(\frac{23}{16}\right)^2 = \frac{529}{256} \approx 2.0664$$

Conclusion: The third-order approximation 1.4375 is closer to $\sqrt{2} \approx 1.414$ and slightly overestimates when squared.

Root Finding: The Newton-Raphson Method

Objective: Given a function $f(x)$, we want to find x such that $f(x) = 0$.
Initial Guess: Start with an initial guess x_0.
Linear Approximation: Approximate $f(x)$ near x_0 using the first two terms of its Taylor series expansion:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

Root Approximation: Set this approximation equal to zero to find the next approximation x_1:

$$0 = f(x_0) + f'(x_0)(x - x_0)$$

Derivation of the Newton-Raphson Iteration

Isolate x: Rearranging the equation from the previous slide to solve for x:

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Iterative Process: Use this formula iteratively to refine the approximation. If x_1 is the next approximation, then:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Convergence: Under certain conditions (e.g., $f(x)$ is continuous and $f'(x) \neq 0$), the sequence x_n converges to the actual root.

Applying Newton-Raphson Method

Procedure:

1. Start with an initial guess x_0.
2. Compute the next approximation using:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

3. Repeat until $|x_{n+1} - x_n|$ is sufficiently small.

Advantages

- Fast convergence near the root.
- Simple iterative formula.

Newton's Method: Interactive Demo

Find the extrema for $f(x) = (x-1)e^x - 16x + 2$

Example.

- Find the extrema for $f(x) = (x-1)e^x - 16x + 2$
- Use the Newton-Raphson method to approximate the solution of the roots for $f'(x)$

Finding Extrema for $f(x) = (x-1)e^x - 16x + 2$

Step 1. Compute the First Derivative

$$f'(x) = D_x(x) = \frac{d}{dx}\left[(x-1)e^x - 16x + 2\right] = xe^x - 16$$

Step 2. Compute the first derivative of $D_x(x)$, i.e., second derivative of $f(x)$

$$D_x'(x) = f''(x) = \frac{d}{dx}\left[xe^x - 16\right] = e^x(x+1)$$

Step 3. Set up the Newton-Raphson Method
To find the extrema, solve for $f'(x) = D_x(x) = 0$ using the Newton-Raphson method:

$$x_{n+1} = x_n - \frac{D_x(x_n)}{D_x'(x_n)} = x_n - \frac{x_n e^{x_n} - 16}{e^{x_n}(x_n+1)}$$

Finding Extrema for $f(x) = (x-1)e^x - 16x + 2$

Numerical solution using Newton's method:

Newton–Raphson for $f'(x) = xe^x - 16$: Interactive Demo